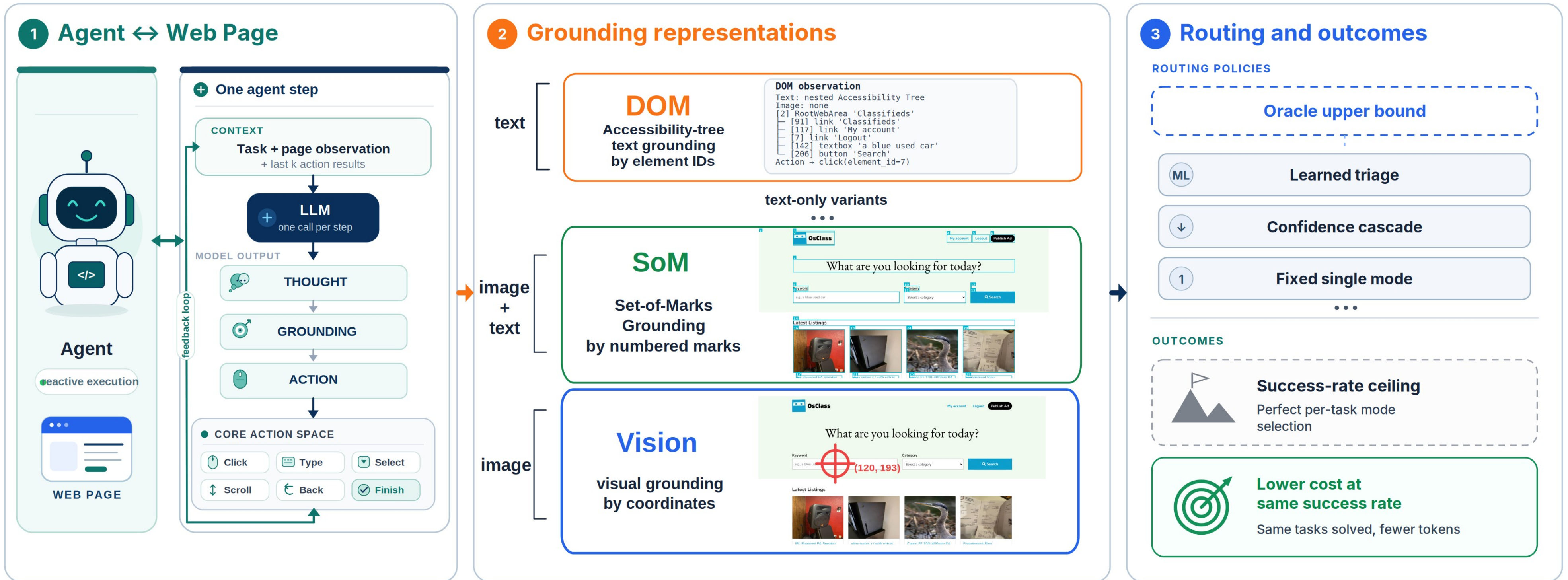


Testing when richer web-agent representations help — and whether their value can be predicted cheaply

WHAT AN AGENT SEES, AND WHO CHOOSES



THE SAME TASK, SEEN TWO WAYS

“Go to my listing of the blue bike and change the price to \$85.50 — and say so in the description.”

READ
page text only
12 steps
8 pages seen
\$0.09 solved

LOOK
screenshot only
26 steps
9 pages seen
\$0.10 gave up

Both start on the same form. Then READ finishes it, while LOOK ends up back on the page it started from. One recorded run per view; both outcomes repeat on an independent rerun.

step 4 · type → \$0.020 spent

step 8 · type → \$0.050 spent

step 10 · click → \$0.086 spent

step 2 · click → \$0.011 spent

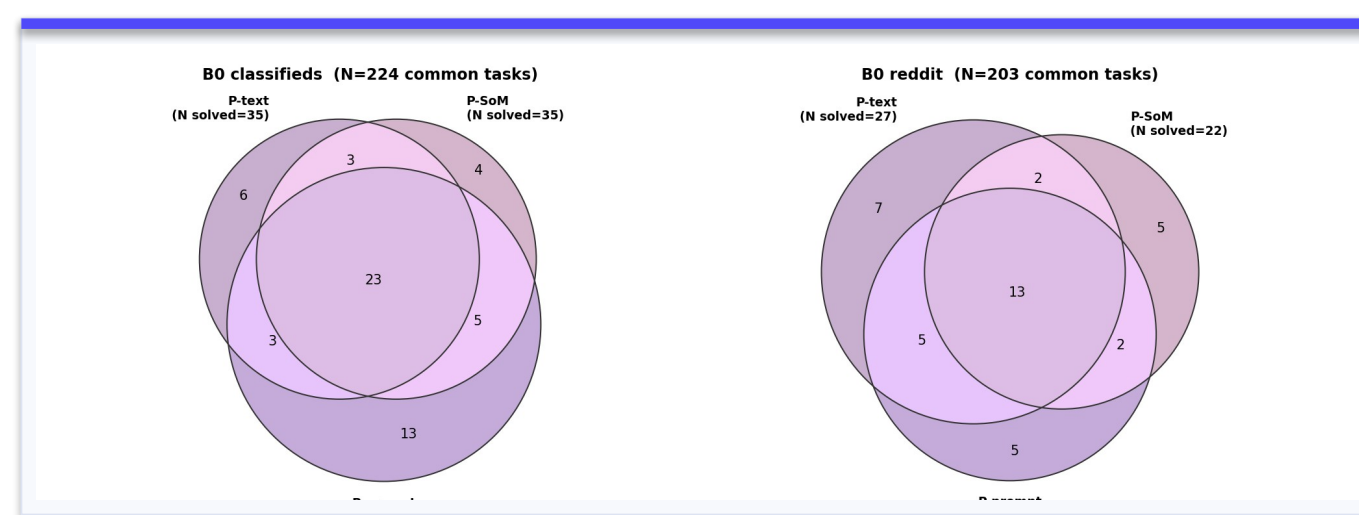
step 17 · click → \$0.071 · left the form

step 19 · click → \$0.079 · back on frame 1

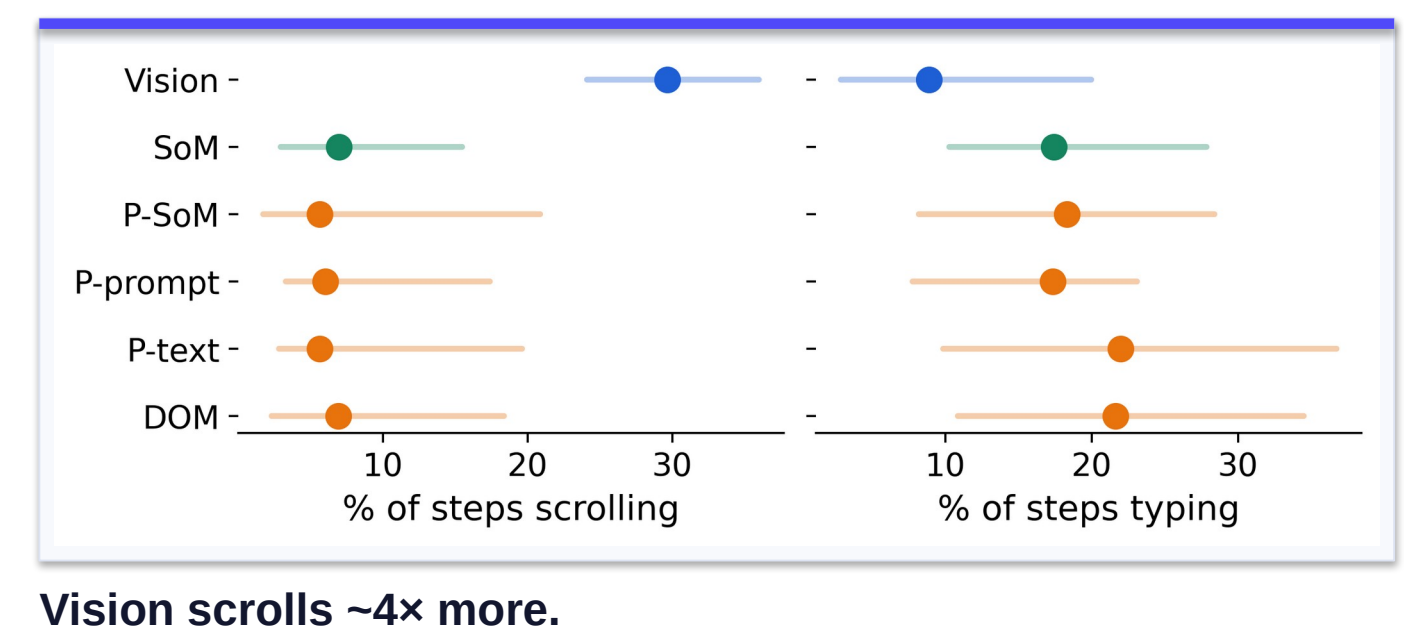
1 · SIX VIEWS, EIGHT SETTINGS

	VisualWebArena						WebArena	
	DOM	P-prompt	P-text	P-SoM	SoM	Vision	wa, red	wa, red
no image	17.4	19.6	15.6	15.6	27.2	25.0	26.9	16.4
text + image	14.3	12.3	13.3	10.8	14.8	7.4	14.3	13.5
image only	6.2	6.7	7.6	6.7	7.4	2.5	12.5	9.6
	5.9	5.4	5.9	5.9	2.2	2.0	19.2	9.6
	1.3	1.8	0.5	0.9	1.0	2.0	19.2	9.6
	3.9	0.0	2.0	0.5	1.0	2.0	19.2	9.6
	26.9	16.4	35.6	25.0	22.1	19.2	16.4	9.6

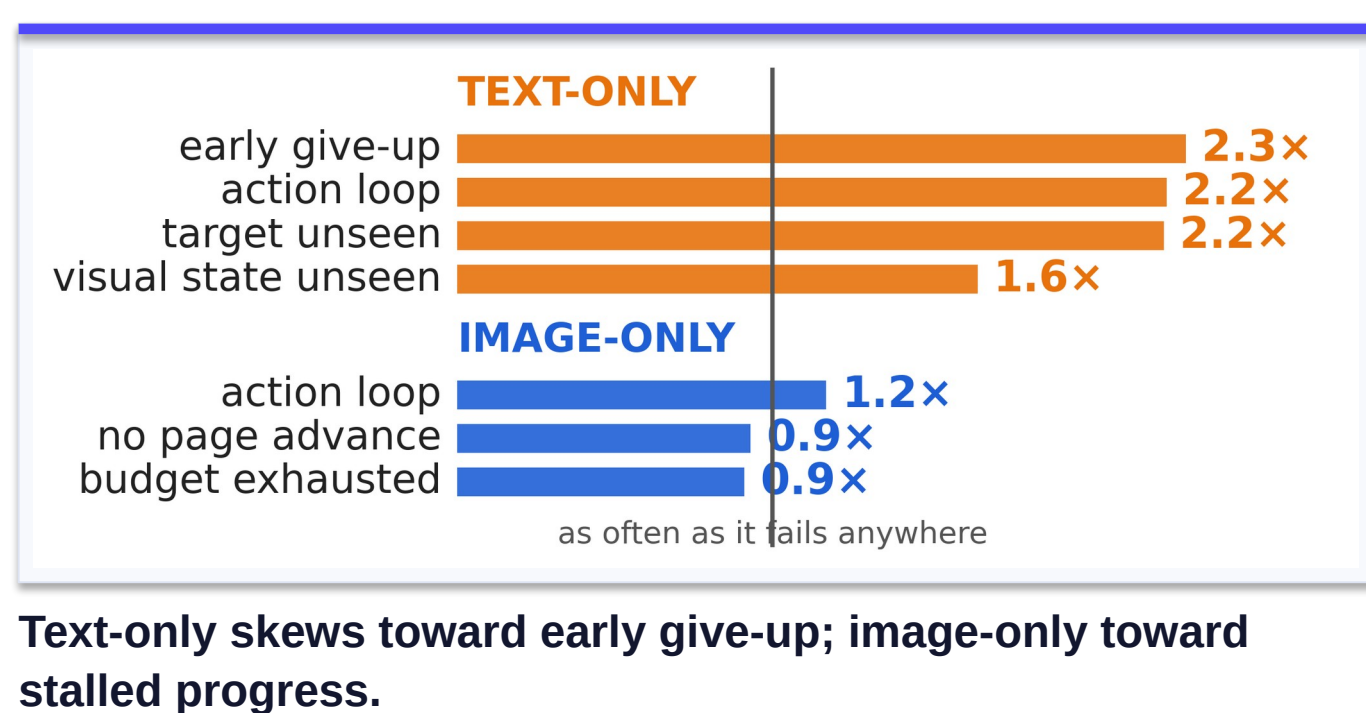
2 · THEY SOLVE DIFFERENT TASKS



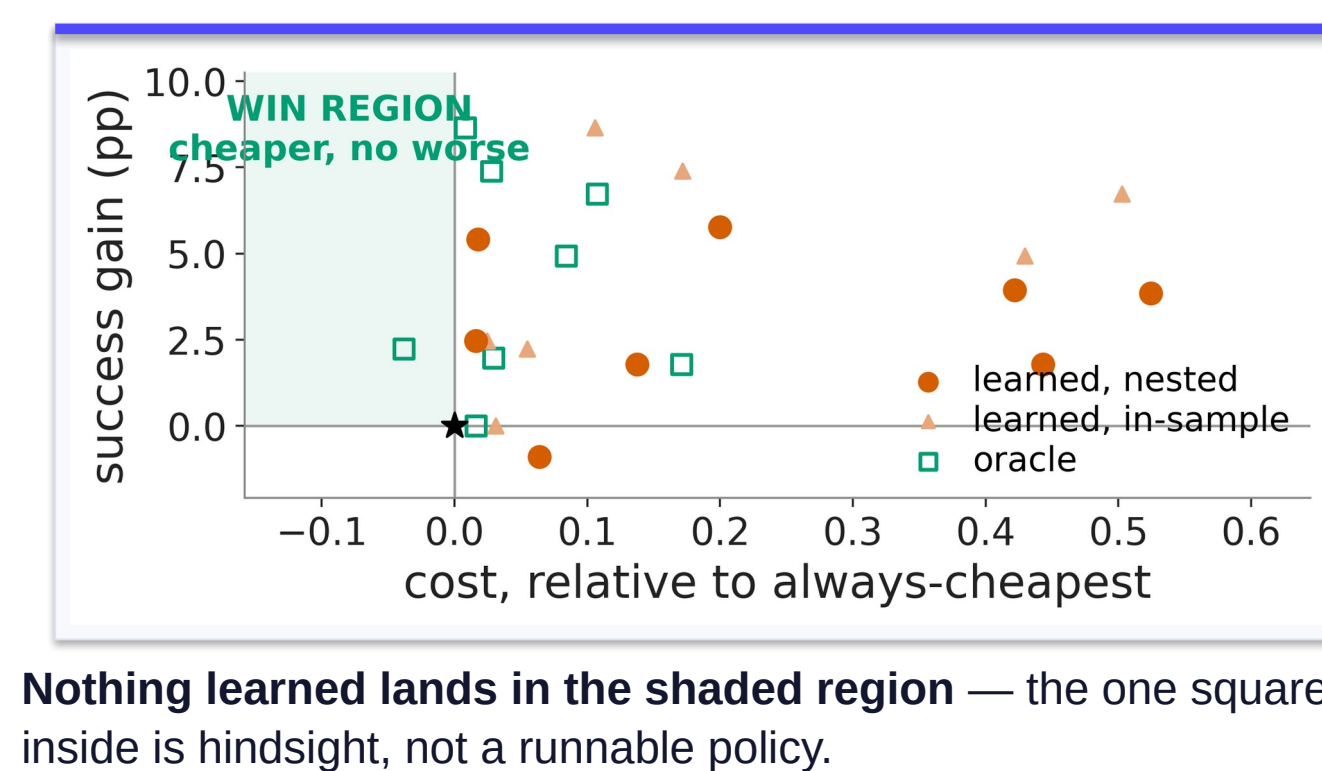
3 · THEY BEHAVE DIFFERENTLY



4 · AND THEY FAIL DIFFERENTLY



5 · SO CHOOSE PER TASK? NOT SO FAST.



6 · AND THIS IS WHY

